

Lecture 1 Overview of Spatial and Temporal Problems

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STP598 Spatiotemporal Analysis
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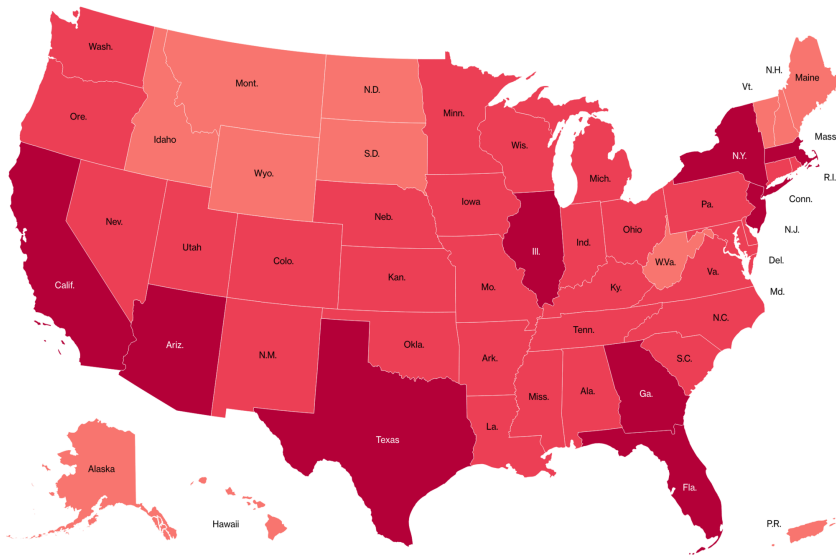
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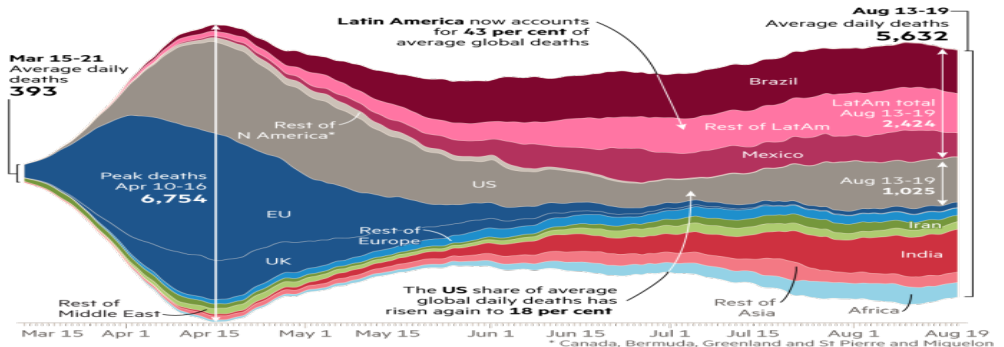
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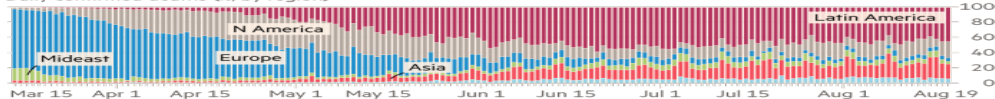
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Resurgence in Covid-19 deaths approaching mid-April peak

Daily deaths of patients diagnosed with coronavirus (7-day rolling average)



Daily confirmed deaths (% by region)



FT graphic: Steven Bernard / @sdbernard

Sources: FT analysis of data from the ECDC, the Covid Tracking Project, the UK Dept of Health and Social Care and the Spanish Ministry of Health

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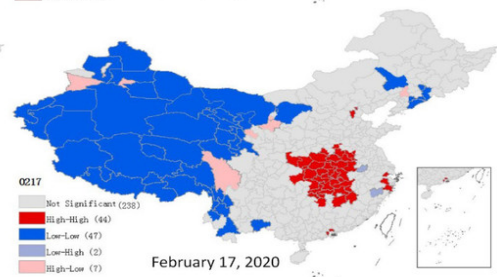
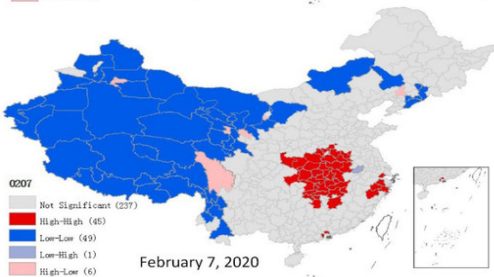
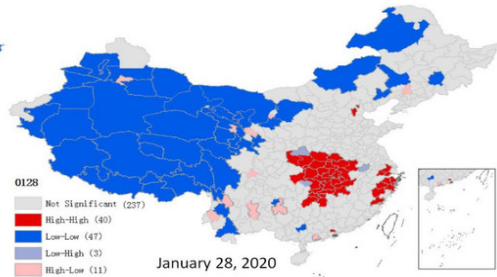
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RISING AND FALLING NEW CORONAVIRUS CASES

CHANGE IN DAILY NUMBER OF NEW CASES

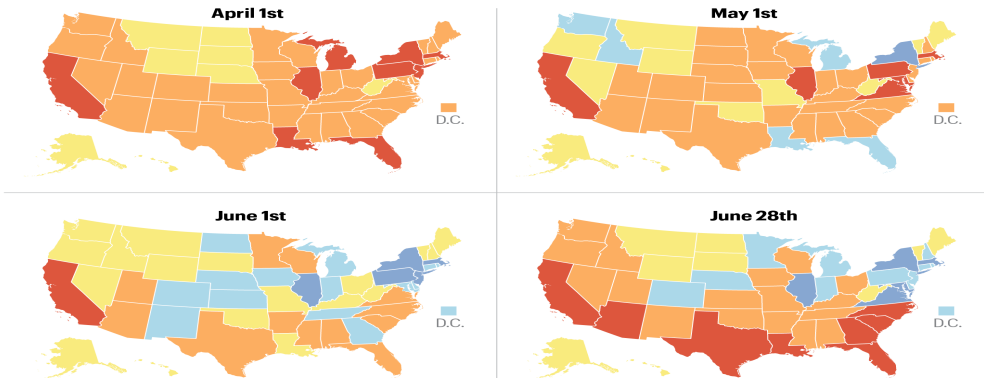
STRONG INCREASE

INCREASE

FLAT

DECREASE

STRONG DECREASE



SEVEN-DAY AVERAGE OF NEW CASES. "STRONG" CHANGE: IN EXCESS OF 500 CASES; "FLAT": +/- 25
 SOURCE: N.Y. TIMES COMPILATION OF STATE AND LOCAL GOVERNMENTS AND HEALTH DEPARTMENTS DATA

FORTUNE

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When analyzing spatial data, researchers in areas such as climatology, ecology, environmental health and real estate marketing are increasingly faced with challenges:

- highly multivariate, with many important predictors and response variables,
- geographically referenced, and often presented as maps, and
- temporally correlated, as in longitudinal or other time series structures.

Following convention, we classify spatial data into one of three basis types:

- *point-referenced* data: $Y(\mathbf{s})$ is a random vector at location $\mathbf{s} \in \mathbb{R}^r$, where $\mathbf{s}$ varies continuously over fixed $D \subset \mathbb{R}^r$ that contains a r -dimensional rectangle of positive volume;
- *areal* data: D is a fixed domain (of regular or irregular shape), but partitioned into a finite number of areal units with well-defined boundaries;
- *point pattern (process)* data: D is random, while $Y(\mathbf{s})$ can simply equal 1 for $\mathbf{s} \in D$ indicating occurrence of the event.

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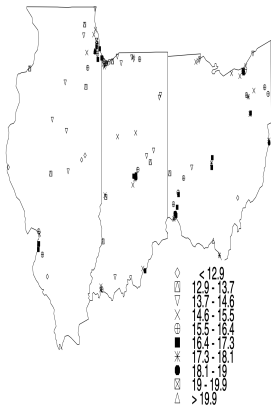


Figure 1.1 Map of PM_{2.5} sampling sites over three midwestern U.S. states; plotting character indicates range of average monitored PM_{2.5} level over the year 2001 (see also color insert).

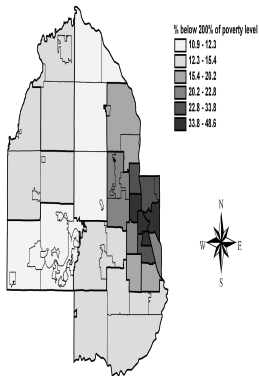


Figure 1.2 ArcView map of percent of surveyed population with household income below 200% of the federal poverty limit, regional survey units in Hennepin County, MN (see also color insert).

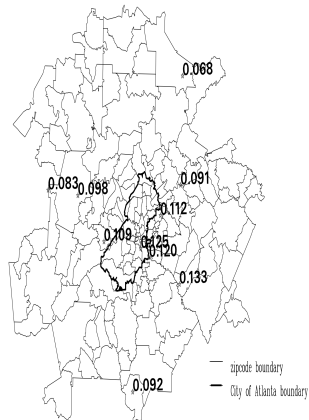


Figure 1.3 Zip code boundaries in the Atlanta metropolitan area and 8-hour maximum ozone levels (ppm) at 10 monitoring sites for July 15, 1995 (see also color insert).

Figure: Spatial Data: Point referenced (left); areal (middle); and point pattern (right)

- Assume the covariance between the random variables at two locations depends on the distance d . One popular association specification is the *exponential* model:

$$\text{Cov}(Y(\mathbf{s}_i), Y(\mathbf{s}_{i'})) = C(d_{ii'}) = \sigma^2 e^{-\phi d_{ii'}} + \tau^2 I(i = i') \quad (1)$$

where $d_{ii'}$ is the distance between sites $\mathbf{s}_i$ and $\mathbf{s}_{i'}$, σ^2 is called *magnitude (partial sill)* parameter, $\phi > 0$ is called *decay* ($1/\phi$ named *range*) parameter, and τ^2 is called *nugget effect*.

- Note $C(d_{ii}) = \text{Var}(Y(\mathbf{s}_i)) = \sigma^2 + \tau^2$ is called *sill*.

- Assume a multivariate normal (Gaussian) distribution for the data $\mathbf{Y} \equiv \{Y(\mathbf{s}_i)\}_{i=1}^n$:

$$\mathbf{Y}|\mu, \boldsymbol{\theta} \sim N_n(\mu, \Sigma(\boldsymbol{\theta})) \quad (2)$$

where μ is the (constant) mean level, and $\Sigma(\boldsymbol{\theta})_{ii'}$ gives the covariance between $Y(\mathbf{s}_i)$ and $Y(\mathbf{s}_{i'})$.

- We denote $\boldsymbol{\theta} = (\tau^2, \sigma^2, \phi)^T$ as nugget, sill and range parameters for the covariance model.
- We use the exponential form for the covariance

$$\Sigma(\boldsymbol{\theta})_{ii'} = \sigma^2 \exp(-\phi d_{ii'}) + \tau^2 I(i = i'), \quad \sigma^2 > 0, \phi > 0, \tau^2 > 0 \quad (3)$$

or powered exponential

$$\Sigma(\boldsymbol{\theta})_{ii'} = \sigma^2 \exp(-\phi d_{ii'}^\kappa) + \tau^2 I(i = i'), \quad \sigma^2 > 0, \phi > 0, \tau^2 > 0, \kappa \in (0, 2] \quad (4)$$

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- For modeling areal data, the geographic regions are divided into blocks (zip codes, counties, etc..), denoted as B_i . The spatial association is defined among *neighborhood* based on the arrangement of the blocks.
- Two popular models that incorporate neighborhood information are the *simultaneously* and *conditionally autoregressive* models (SAR and CAR), originally developed by Whittle (1954) and Besag (1974) respectively.
- SAR model is computationally convenient for use with likelihood methods.
- CAR is computationally convenient for Gibbs sampling for Bayesian model fitting.

- CAR model incorporates spatial correlation through a vector of spatially varying random effects $\phi = (\phi_1, \dots, \phi_n)^T$.
- Let $Y_i \equiv Y(B_i) \stackrel{iid}{\sim} N(\phi_i, \sigma^2)$. CAR model imposes

$$\phi_i | \phi_{(-i)} \sim N \left(\mu + \sum_{j=1}^n a_{ij}(\phi_j - \mu), \tau_i^2 \right) \quad (5)$$

where $\phi_{-i} = \{\phi_j : j \neq i\}$, τ^2 is the conditional variance, and a_{ij} are known or unknown constants such that $a_{ii} = 0$ for $i = 1, \dots, n$.

- Denote $A = (a_{ij})$, $M = \text{diag}(\tau_1^2, \dots, \tau_n^2)$. By Brook's Lemma,

$$\phi(\phi) \propto \exp\{-(\phi - \mu \mathbf{1})^T M^{-1}(I - A)(\phi - \mu \mathbf{1})/2\} \quad (6)$$

- A common way to construct A and M is to let $A = \rho W$ and $M^{-} = \tau^{-2} \text{diag}(w_{i+})$, where $w_{i+} = \sum_j w_{ij}$.
- ρ is referred to as *spatial correlation* parameter, and $W = (w_{ij})$ is a neighborhood matrix for areal units with

$$w_{ij} = \begin{cases} 1 & \text{if subregions } i \text{ and } j \neq i \text{ share boundary} \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

- Denote $\alpha = (\rho, \tau^2)$. The covariance matrix of ϕ becomes

$$C(\alpha) = \tau^2 [\text{diag}(w_{i+}) - \rho W]^{-1} \quad (8)$$

where the inverse exists for an appropriate range of ρ .

- Special case (assume $\sum_i \phi_i = 0$):

$$\phi_i | \phi_{(-i)} \sim N(\bar{\phi}_i, \tau^2 / m_i) \quad (9)$$

where $\bar{\phi}_i$ is the average of $\phi_{j \neq i}$ adjacent to ϕ_i , and m_i is the degree of ϕ_i .

- In the point process model, the spatial domain D is random so that the elements of D are locations of random events that constitute the spatial point pattern.
- $Y(\mathbf{s}) \equiv 1$ for all $\mathbf{s} \in D$. Question of interest (QoI) is whether the data are *clustered* or scatter by chance.
- The underlying process is usually modeled as a *homogeneous Poisson process*, which implies the expected number of occurrences in region A is $\lambda|A|$.
- The most common statistics for measuring clustering is *Ripley's K function*:

$$K(d) = \frac{1}{\lambda} E[\text{number of points within } d \text{ of an arbitrary point}] \quad (10)$$

- Theoretic uniform distribution: $K(d) = \pi d^2$; clustered: $K(d) > \pi d^2$; scattering: $K(d) < \pi d^2$.
- In practice, we could estimate K by the following estimator

$$\hat{K}(d) = n^{-2}|A| \sum_{i \neq j} p_{ij}^{-1} I_d(d_{ij}) \quad (11)$$

where n is the number of points in A , d_{ij} is the distance between points i and j , p_{ij} is the proportion of the circles with center i and passing through j that lies within A , and $I_d(d_{ij})$ equals 1 if $d_{ij} < d$ and 0 otherwise.

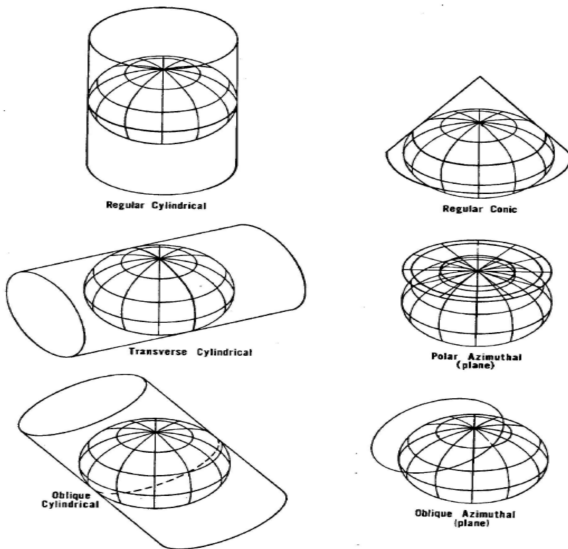


Figure 1.4 The geometric constructions of projections using developable surfaces 19/41

- Given a sphere with geographical coordinate system (λ, ϕ) for longitude and latitude, construct an appropriate (rectangular or polar) coordinate system (x, y) such that

$$x = f(\lambda, \phi), \quad y = g(\lambda, \phi) \quad (12)$$

We want to find f, g such that it preserves area.

- Consider infinitesimal quadrilateral, $ABCD$, given by the vertices,

$$A = (\lambda, \phi), \quad B = (\lambda, \phi + d\phi), \quad C = (\lambda + d\lambda, \phi), \quad D = (\lambda + d\lambda, \phi + d\phi) \quad (13)$$

- Under the above mapping, we have another infinitesimal quadrilateral, $A'B'C'D'$, given by

$$\begin{aligned} A' &= (f(\lambda, \phi), g(\lambda, \phi)), & B' &= (f(\lambda, \phi + d\phi), g(\lambda, \phi + d\phi)), \\ C' &= (f(\lambda + d\lambda, \phi), g(\lambda + d\lambda, \phi)), & D' &= (f(\lambda + d\lambda, \phi + d\phi), g(\lambda + d\lambda, \phi + d\phi)) \end{aligned} \quad (14)$$

- To force their areas equal, we set

$$|ABCD| = R \cos \phi d\lambda R d\phi = |A'B'C'D'| = \left(\frac{\partial f}{\partial \lambda} \frac{\partial g}{\partial \phi} - \frac{\partial f}{\partial \phi} \frac{\partial g}{\partial \lambda} \right) d\lambda d\phi \quad (15)$$

- We obtain the following PDE for area-preserving projections

$$\frac{\partial f}{\partial \lambda} \frac{\partial g}{\partial \phi} - \frac{\partial f}{\partial \phi} \frac{\partial g}{\partial \lambda} = R^2 \cos \phi \quad (16)$$

- *Sinusoidal* projection: $f(\lambda, \phi) = R\lambda \cos \phi$; $g(\lambda, \phi) = R\phi$.
- *Lambert cylindrical* projection: $f(\lambda, \phi) = R\lambda$; $g(\lambda, \phi) = R \sin \phi$.

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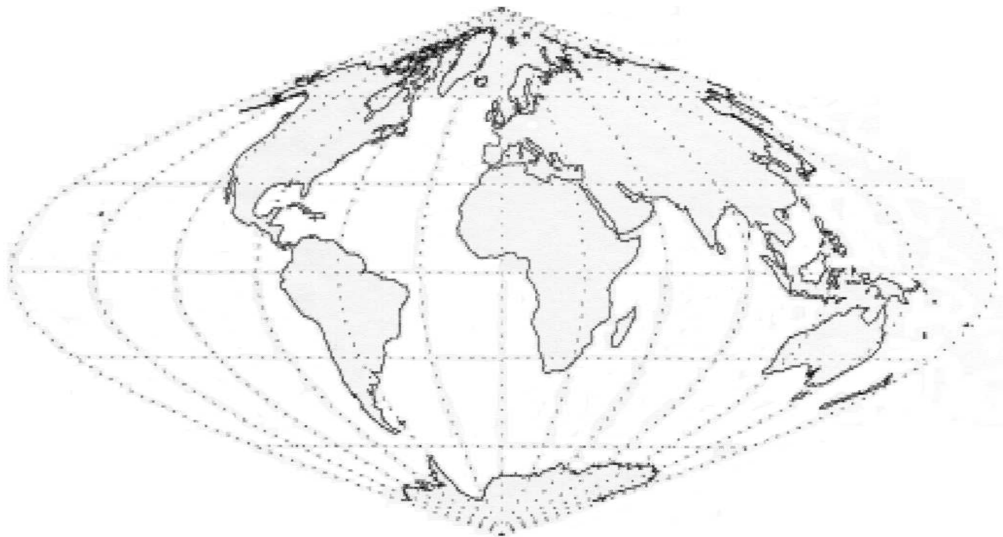


Figure 1.5 *The sinusoidal projection.*

- Conformal projections force their angles equal, i.e. $\angle(AB, AC) = \frac{\pi}{2} = \angle(A'B', A'C')$, which implies

$$\cos \angle(A'B', A'C') = \frac{\partial f}{\partial \lambda} \frac{\partial f}{\partial \phi} + \frac{\partial g}{\partial \lambda} \frac{\partial g}{\partial \phi} = 0 \quad (17)$$

- One simple sufficient condition yields the following PDE:

$$\frac{\partial f}{\partial \lambda} = \frac{\partial g}{\partial \phi} \cos \phi; \quad \frac{\partial g}{\partial \lambda} = -\frac{\partial f}{\partial \phi} \cos \phi \quad (18)$$

- Mercator* projection: $f(\lambda, \phi) = R\lambda$; $g(\lambda, \phi) = R \log \tan \left(\frac{\pi}{4} + \frac{\phi}{2} \right)$.

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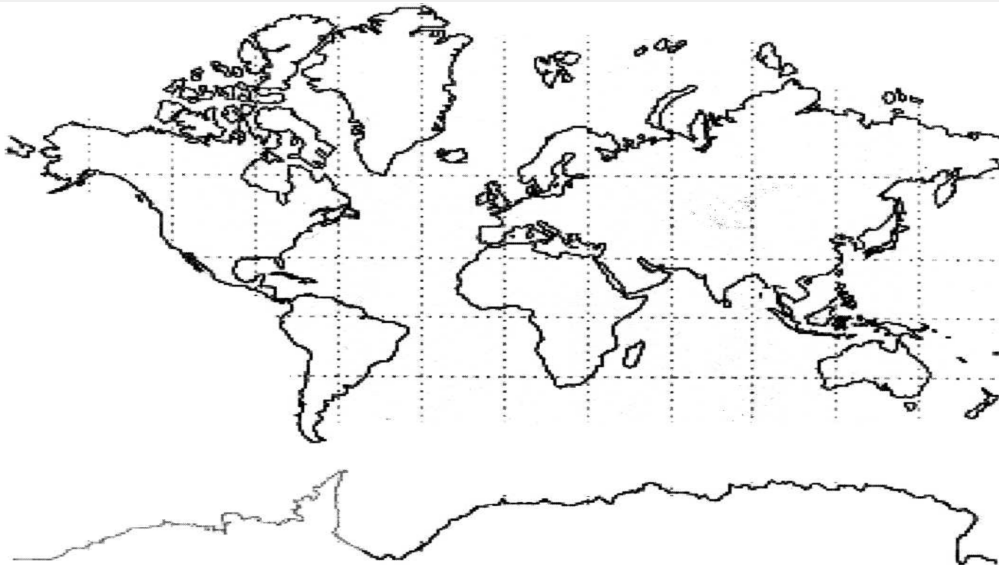


Figure 1.6 *The Mercator projection.*

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- Given two points on the surface of the earth $P_1 = (\lambda_1, \theta_1)$ and $P_2 = (\lambda_2, \theta_2)$. How to compute their distance?
- We have the following relationship between Euclidean coordinate and spherical (longitude-latitude) coordinate

$$\begin{aligned}x &= R \cos \theta \cos \lambda \\y &= R \cos \theta \sin \lambda \\z &= R \sin \theta\end{aligned}\tag{19}$$

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When analyzing time series data, researchers in areas such as economics, climatology, epidemiology, and neuroscience are increasingly faced with challenges:

- highly multivariate, with many important predictors and response variables,
- non-stationary, hard to predict,
- often having single history, or missing data, and
- spatially correlated, as in multi-site signals or other spatially dependent multivariate data.

There are two separate, but not necessarily mutually exclusive methods for time series analysis:

- *time domain* approach views the investigation of lagged relationships as most important (e.g., how does what happened today affect what will happen tomorrow).
- *frequency domain* approach views the investigation of cycles as most important (e.g., what is the economic cycle through periods of expansion and recession).

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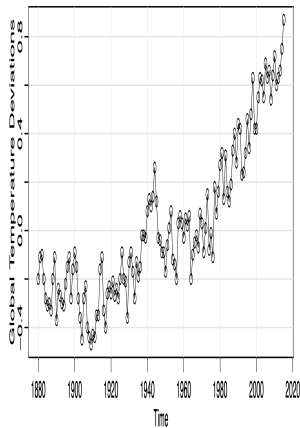


Fig. 1.2. Yearly average global temperature deviations (1880–2015) in degrees centigrade.

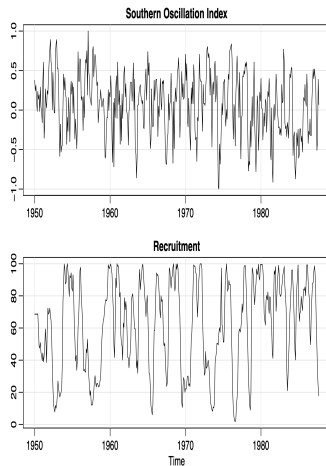


Fig. 1.5. Monthly SOI and Recruitment (estimated new fish), 1950–1987.

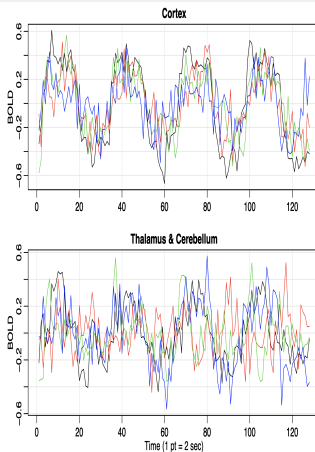


Fig. 1.6. fMRI data from various locations in the cortex, thalamus, and cerebellum; $n = 128$ points, one observation taken every 2 seconds.

Figure: Time Series Data: Non-stationary (left); cyclic (middle); and multivariate (right)

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- We consider a time series as a sequence of random variables, $x_1, x_2, x_3, \dots$, denoted as $\{x_t\}$, indicating random value at time t .
- The collection of random variables $\{x_t\}$ is called a *stochastic process*. The observed values of a stochastic process is termed a *realization*. *Time series* $\{x_t\}$ is generically referred to as the process or a particular realization. How to model it?
- We could model x_t as a linear combination of *white noise* $\{w_t\}$, hence named *moving average model* **MA**(q):

$$x_t = \theta(B)w_t, \quad \theta(B) = \sum_{i=0}^q \theta_i B^i, \quad w_t \sim wn(0, \sigma_w^2) \quad (20)$$

where B is the backward operator such that $B^i w_t = w_{t-i}$.

- Or we could model x_t as a linear combination of its history, hence named autoregressive model **AR**(p):

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t \quad (21)$$

Therefore it can be written as

$$\phi(B)x_t = w_t, \quad \phi(B) = 1 - \sum_{i=1}^p \phi_i B^i \quad (22)$$

- Or combining moving average and autoregression to obtain **ARMA**(p, q) model:

$$\phi(B)x_t = \theta(B)w_t \quad (23)$$

- A general model that is of interest as linear regression is the state-space model introduced by Kalman (1960) and Kalman and Bucy (1961).
- The model arose in control engineering where a state equation governs the motion of x_t and the data y_t reflect the information that can be observed.
- In general, there are two principles assumed:
 - 1 The state process $\{x_t\}$ is Markovian, i.e. the future $x_s : s > t$ and the past $x_s : s < t$ are independent conditioned on the present x_t ;
 - 2 The observations $\{y_t\}$ are independent given the states x_t .

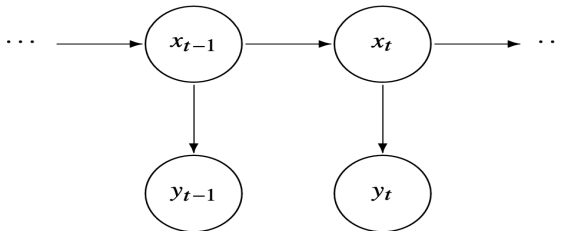


Fig. 6.1. *Diagram of a state space model.*

- The linear Gaussian model is the most fundamental state-space model which takes the following form:

$$\begin{aligned}x_t &= \Phi x_{t-1} + w_t, & w_t &\stackrel{iid}{\sim} N_p(0, Q) \\ y_t &= A_t x_t + v_t, & v_t &\stackrel{iid}{\sim} N_q(0, R)\end{aligned}\tag{24}$$

where we assume $x_0 \sim N_p(\mu_0, \Sigma_0)$, and A_t is a $q \times p$ measurement/observation matrix.

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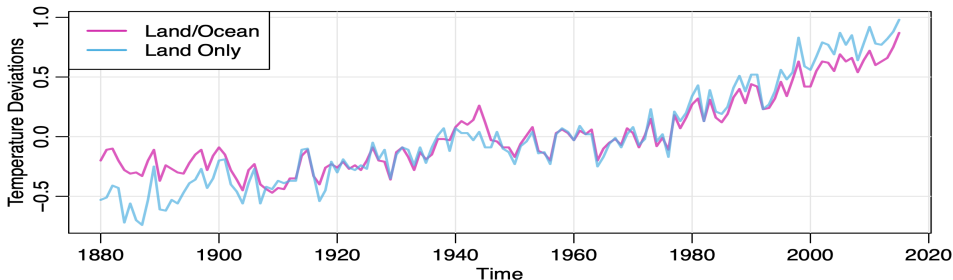
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Fig. 6.3. Annual global temperature deviation series, measured in degrees centigrade, 1880–2015. The series differ by whether or not ocean data is included.

- Two time series of temperature $\{y_{t1}, y_{t2}\}$ are modeled as a linear Gaussian state-space model

$$\begin{bmatrix} y_{t1} \\ y_{t2} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} x_t + \begin{bmatrix} v_{t1} \\ v_{t2} \end{bmatrix}, \quad R = \text{Var} \begin{bmatrix} v_{t1} \\ v_{t2} \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \quad (25)$$

$$x_t = \delta + x_{t-1} + w_t, \quad Q = \text{Var}(w_t)$$

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- The objective of the analysis involving the state-space model is to produce estimators for the underlying unobserved (latent) process x_t , given the data $y_{1:s} = \{y_1, \dots, y_s\}$ up to time s .
- Based on the relationship between t and s , the problems can be divided into 3 types:
 - ① *smoothing*: $t < s$
 - ② *filtering*: $t = s$
 - ③ *forecasting or prediction*: $t > s$

- *Hidden Markov models (HMM)* were developed parallel to state-space models in Goldfeld and Quandt (1973) and Lindgren (1978). A good summary is Rabiner and Juang (1986).
- Recall the first principle of state-space model is latent process x_t being Markovian. If the states x_t are a discrete valued Markov chain, then the state-space model is called *hidden Markov model (HMM)*.
- Assume the states, x_t , are Markov chain taking values in $1, \dots, m$, with the following stationary distribution and transition probabilities

$$\pi_j = \Pr(x_t = j), \quad \pi_{ij} = \Pr(x_{t+1} = j | x_t = i) \quad (26)$$

for $t = 0, 1, 2, \dots$ and $i, j = 1, \dots, m$.

- We then need to specify the observation model:

$$p_j(y_t) = \Pr(y_t | x_t = j) \quad (27)$$

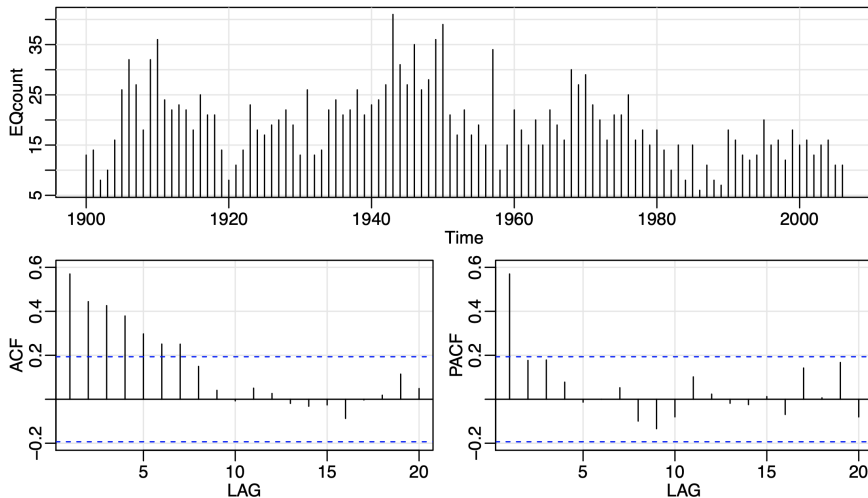


Fig. 6.12. Top: Series of annual counts of major earthquakes (magnitude 7 and above) in the world between 1900–2006. Bottom: Sample ACF and PACF of the counts.

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Map projections

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Autoregressive Moving

Average Models

State Space Models

Spatiotemporal
Problems

When analyzing spatiotemporal data, researchers in various areas are faced with challenges:

- highly multivariate, with many important predictors and response variables,
- often having single history, or missing data, and
- intricate relationship between space and time.

- Let $Y(\mathbf{s}, t)$ denote the measurement at location $\mathbf{s}$ at time t . We adopt the following general form

$$Y(\mathbf{s}, t) = \mu(\mathbf{s}, t) + e(\mathbf{s}, t) \quad (28)$$

where $\mu(\mathbf{s}, t)$ denotes the mean structure and $e(\mathbf{s}, t)$ denotes the residual.

- If $\mathbf{x}(\mathbf{s}, t)$ is a vector of covariates associated with $Y(\mathbf{s}, t)$, then we could model $\mu(\mathbf{s}, t) = \mathbf{x}(\mathbf{s}, t)\beta(\mathbf{s}, t)$.
- The error process $e(\mathbf{s}, t)$ is usually modeled with further details (Gelfand, Ecker, Knight, and Sirmans, 2003):

$$e(\mathbf{s}, t) = \alpha(t) + w(\mathbf{s}) + \epsilon(\mathbf{s}, t) \quad (29)$$

$$e(\mathbf{s}, t) = \alpha_{\mathbf{s}}(t) + \epsilon(\mathbf{s}, t) \quad (30)$$

$$e(\mathbf{s}, t) = w_t(\mathbf{s}) + \epsilon(\mathbf{s}, t) \quad (31)$$

where $\epsilon(\mathbf{s}, t) \stackrel{iid}{\sim} N(0, \sigma_{\epsilon}^2)$.

Overview

S.Lan

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Spatiotemporal Problems

- How do we characterize spatiotemporal correlation?
- How do we deal with sparse/missing data?
- How do we explain response changes with covariates?
- ...?